FACULTY O ND COMPUTER SCIENCE
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DEPARTMENT OF MECHATRONICS ENGINEERING

CCE 220L : Digital Systems LAB

2 FLOORS ELEVATOR SYSTEM

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Introduction

We are representing our project which is a 2-floor elevator system. Our goal was to design a simple digital system that controls the movement of the elevator between two floors.

WHY	WHERE	HOW
To demonstrate how real-world electromechanical systems can be reduced to elegant digital logic circuits using minimized Boolean expressions.	Applicable in building automation, industrial control systems, and embedded systems — any environment requiring floor-based navigation.	Using logic gates (AND, NOT), D flip-flops for state storage, and K-map minimization — implemented and verified on a breadboard.

Inputs, Outputs & State Diagram

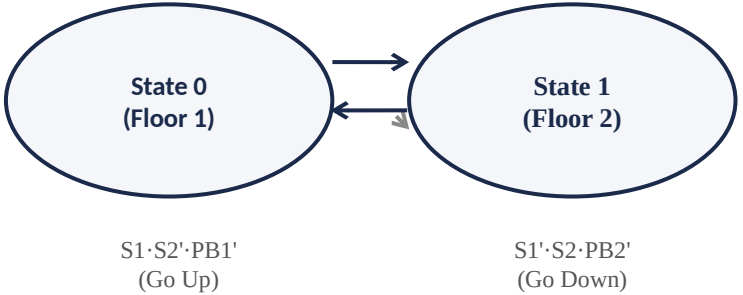
System Inputs

- S1 : switch for floor 1 call
- S2 : switch for floor 2 call
- PB1 : floor 1 position sensor (push button)
- PB2 : floor 2 position sensor (push button)

System Outputs

- Q1 : Control elevator direction
- Q2 : Control elevator direction
- Q1= 0, Q2=1 → Motor moves UP (Floor 2)
- Q1=1, Q2=0 → Motor moves DOWN (Floor 1)
- Q1 = 0, Q2=0 → Motor STOPPED

State Diagram



State Storage: D Flip-Flops

- D flip-flops control the motor
- Outputs Q1 and Q2 control the motor direction
- Transitions triggered by clock edge on floor arrival

Truth Table

S1	S2	PB1	PB2	Q1	Q2	Q1*	Q2*	D1	D2
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	1	0	1
0	1	0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	1	0	1
0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	0	1	0	1	0
1	0	0	1	0	0	1	0	1	0
1	0	1	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0

Motor Control Logic

Q1	Q2	Motor
0	0	OFF
0	1	Clockwise
1	0	Counter-CW
1	1	OFF

K-Maps & Boolean Functions

K-Map for D1

S1·S2 \ PB1·PB2	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	1	0	0

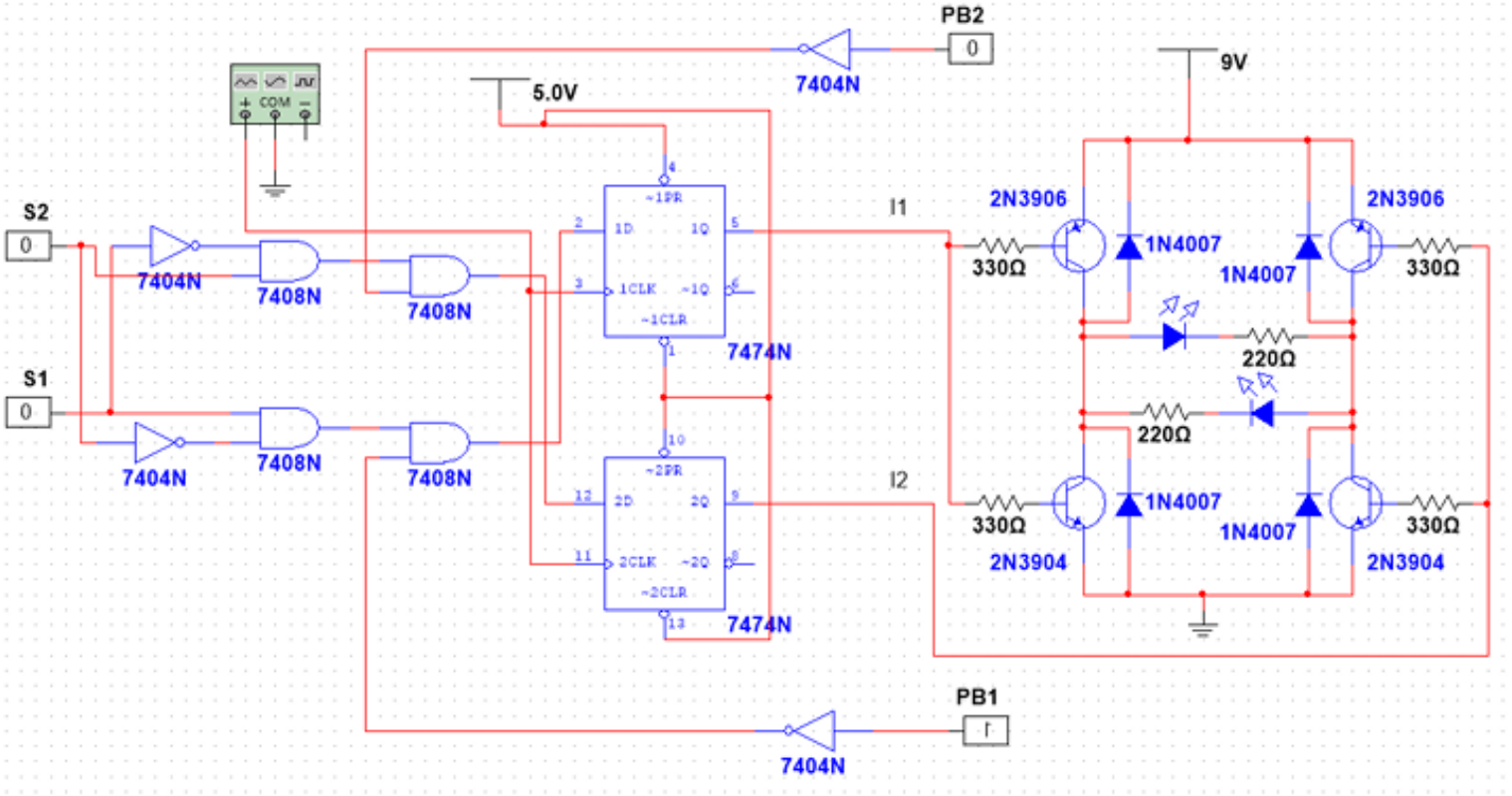
D1 = S1 · S2' · PB1'

K-Map for D2

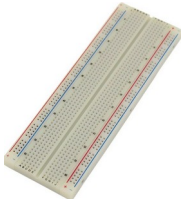
S1·S2 \ PB1·PB2	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	0	0	0	0
10	0	0	0	0

D2 = S1' · S2 · PB2'

Circuit Diagram



Materials Used



Breadboard



Logic gates ICs



Connecting wires



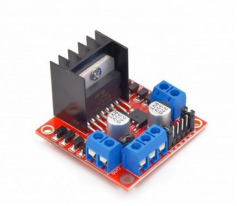
D Flip-Flops



Push buttons



DC Motor



H-Bridge

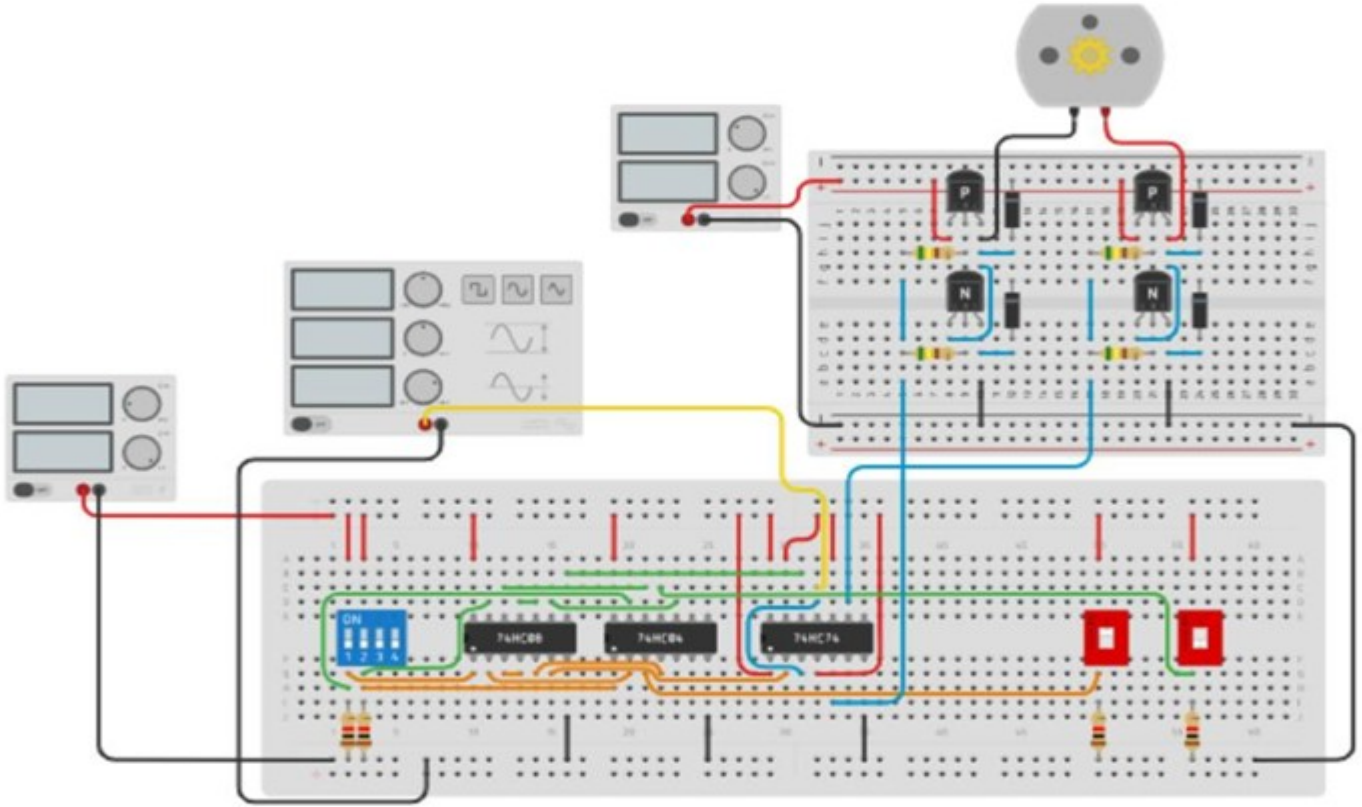


LEDs

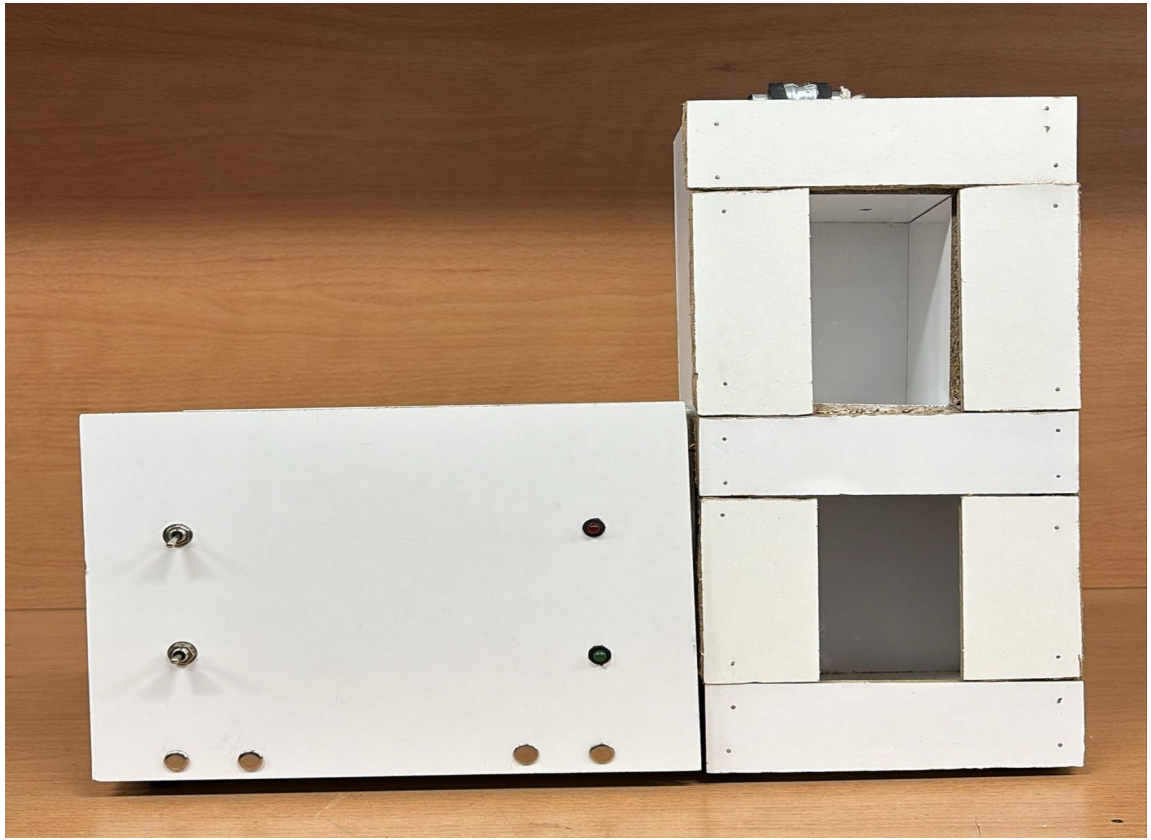


Resistors

Implementation

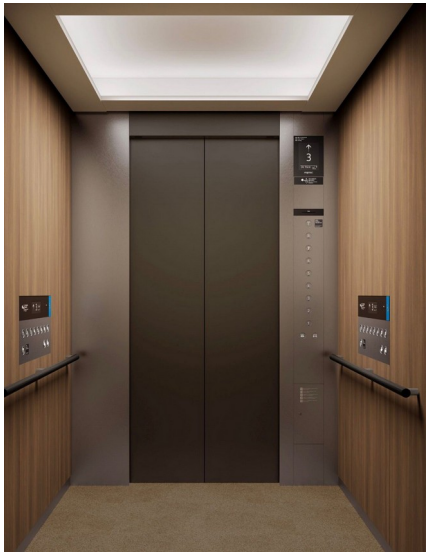


Design



Future work

- Our objective was to create a 2-floor digital elevator system capable of receiving floor requests and controlling the motor to move the elevator cabinet up and down and where to stop.
- Since we have successfully designed a 2 floors elevator we can also increase the number of floors instead of limiting it to only 2 floors.
- Additional sensors , floor indicators, automatic doors, alarms, and emergency stop functions can be added to improve the safety and accuracy of the elevator movement.



Conclusion

- Designed and implemented a two-floor elevator system using digital logic.
- Successfully tested the system with all input combinations producing correct outputs.
- Used K-map minimization to simplify the circuit design.
- Built the system using basic logic gates and D flip-flops.
- Demonstrated how digital logic can solve real-world control problems.
- Implemented the project as a working breadboard prototype.

Thank you

Q&A